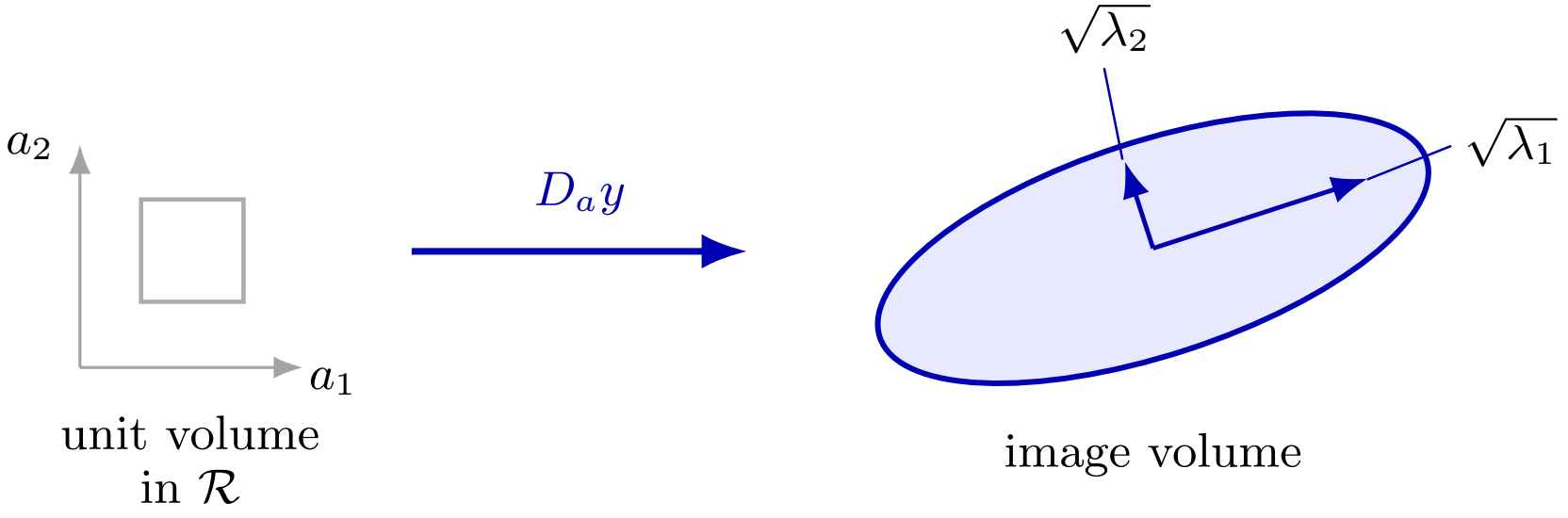
Retained coordinate Jacobian

$$a \in \mathcal{R}, \qquad y(a) = N^{-1/2} W_{\mathcal{R}} a \qquad D_a y := \frac{\partial y}{\partial a} = N^{-1/2} W_{\mathcal{R}}$$



unit volume in $\mathcal{R}$

image volume

$$X_{\mathcal{R}} = (D_a y)^T (D_a y)$$

$$J^W_{\mathcal{R}} = \frac{d\mathrm{Vol}(y)}{d\mathrm{Vol}(a)} = \sqrt{\det X_{\mathcal{R}}}$$

$$\log J^W_{\mathcal{R}} = \frac{1}{2} \operatorname{Tr} \log X_{\mathcal{R}}$$

$$J^{\mathrm{rel}}_{\mathcal{R}} = 1 \iff \operatorname{Tr} \log \widetilde{X}_{\mathcal{R}} = 0$$